

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 314 — Past Questions & Materials  
**by Renance**

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Course: MAR 314  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 3  
Levels: 300

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## Paper 1 of 3 — MAR 314 Marine Electrical Technology

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: FIRST SEMESTER COURSE CODE: MAR 314

SESSION: 2024/2925 & ah COURSE TITLE: MARINE ELECTRICAL TECHNOLOGY

DATE: 23/05/25 Sa TIME ALLOWED: 2HRS

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

1a). Discuss the notable four conditions of a passenger ship enroute Kawasaki Tapan, «'~1 1s influenced by Fectricel

Systems Onboard it (17 marks)

b) With refererce te ship electrical technology, explain ship primary essential services and ship secondary essential

services. Outline the two examples of primary and secondary essential services. {5.5 marks}

2). In a container ship, two generators J and K are sharing load in parallel. The Real/True power, Reactive power and

power factors of each generator is illustrated' in the table below.

Parameters i Generator J Generator K Total bus bar Loading

True/Real Power 700 KW 650 KW ?

Apparent Power ? ? ? \_

Reactive Power 2 «\_ 600 KVAR 2 |

Power Factor 0.7 lag ? ?

Calculate the following: i). The reactive power of Generator J. [3.5 marks] ii). The apparent power of Generator J. {3.5

te orks] iii). The apperent power of Generator K. [3.5 marks] iv). The power factor of Generator K. [3.5 marks] v). The

a setal-bus bar loadisa i KW, KVAR and power factor. [3.5 marks]

3). An oil tanker, has a 440V, 3-ph, induction Motor The-induction motor has 6 poles and operates at 0.7 power factor

lag and 6% ship. The motor drives a 15KW pump at 38.8 rev/s. The stator and mechanicatlosses.are 750W and 480W

respectively. Calculate: 7

i) The supply frequency. [4 marks] ii). The frequency of the rotor current. {4.5 marks] iii). The power input to the

motor, [45 merks) iv). The current taken from the supply. [4.5 marks]

40). Tn a war ship, a generator onboard has 60 conductors/ph and is needed to generate 500 volts/phase. Calculate the

following: i). The needed flux per pole [2.5 marks} ii), The generator line voltage if the three stator winding are star

connected, [2.5 marks]

b). A long shunt compound generator delivers a load current of 60A at 600V, has armature, series field and shunt field

resistances of 0.07 ohms, 0.04 ohms and 350 ohms respectively. Calculate the generated voltage and the armature

current. Allow 1V per brush for contact drop. [6 marks]

c). Describe in details, field, armature, prime mover, rotor, stator and slip rings as components of an AC generator, that

is needed for onboard operations. [6.5 marks]

5a). Given that the generator  $E_{MF} = 2.22 \text{ } \phi \text{ } \text{V/ph}$ . Use engineering judgement from first principle to prove it. [6.5

marks] ;

b). Explain theory of operations of an AC generator. Also

and impedance of the generator, hysteresis and mechanical losses in AC generator. [5.5 marks]

explain internal voltage drops due to the internal resistance

c). Describe three conditions that must be met prior to synchronization of AC generators for offshore operations. (5.5

marks]

6a). Suppose a cargo ship needs 4 motors for various operations. Sketch/Draw a ship electrical system diagram that

shows how 660V, 440V, 440V and 220V are supplied to various motors onboard the cargo ship. [8 marks]

b). Define load analysis of ship electrical system. [2 marks]

c). List five electrical diagrams used in ship electrical system design. State their advantages. [7.5 marks]

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# Paper 2 of 3 — MAR 314 Marine Electrical Technology 2024/2025

Level 300 • First Semester • PastQuestion

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SESSION: 2024/2025  
DATE: 21/05/2025

COURSE TITLE: MARINE ELECTRICAL TECHNOLOGY  
TIME ALLOWED: 2HRS

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

1a) Discuss the notable four condition of that is influenced by Electric and Magnetic field and its influenced by

b) With reference in ship electrical technology, explain ship primary essential services and ship secondary essential services. Outline the two examples of primary and secondary essential services. [5.5 marks]

2) In a container ship, two generators J and K are sharing load in parallel. The Real/True power, Reactive power and power factors of each generator is illustrated in the table below.

| Parameters      | Generator J | Generator K | Total bus bar Loading |
|-----------------|-------------|-------------|-----------------------|
| True/Real Power | 700 KW      | 650 KW      | ?                     |
| Apparent Power  | ?           | ?           | ?                     |
| Reactive Power  | ?           | 600 KVAR    | ?                     |
| Power Factor    | 0.7 lag     | ?           | ?                     |

Calculate the following: i). The reactive power of Generator J. [3.5 marks] ii). The apparent power of Generator J. [3.5 marks] iii). The apparent power of Generator K. [3.5 marks] iv). The power factor of Generator K. [3.5 marks] v). The total bus bar loading in KW, KVAR and power factor. [3.5 marks]

3). An oil tanker, has a 440V, 3-ph, induction motor. The induction motor has 6 poles and operates at 0.7 power factor lag and 6% slip. The motor drives a 15KW pump at 38.8 rev/s. The stator and mechanical losses are 750W and 480W respectively. Calculate:

i). The supply frequency. [4 marks] ii). The frequency of the rotor current. [4.5 marks] iii). The power input to the motor. [4.5 marks] iv). The current taken from the supply. [4.5 marks]

4a). In a war ship, a generator onboard has 60 conductors/ph and is needed to generate 500 volts/phase. Calculate the following: i). The needed flux per pole [2.5 marks] ii). The generator line voltage if the three stator winding are star connected. [2.5 marks]

b). A long shunt compound generator delivers a load current of 60A at 600V, has armature, series field and shunt field resistances of 0.07 ohms, 0.04 ohms and 350 ohms respectively. Calculate the generate voltage and the armature current. Allow 1V per brush for contact drop. [6 marks]

c). Describe in details, field, armature, prime mover, rotor, stator and slip rings as components of an AC generator, that is needed for onboard operations. [6.5 marks]

5a). Given that the generator E.M.F =  $2.22 \phi$  f z volts/ph. Use engineering judgement from first principle to prove it. [6.5 marks]

b). Explain theory of operations of an AC generator. Also explain Internal voltage drops due to the internal resistance and impedance of the generator, hysteresis and mechanical losses in AC generator. [5.5 marks]

c). Describe three conditions that must be met prior to synchronization of AC generators for offshore operations. [5.5 marks]

6a). Supposed a cargo ship needs 4 motors for various operations. Sketch/Draw a ship electrical system diagram that shows how 660V, 440V, 440V and 220V are supplied to various motors onboard the cargo ship. [8 marks]

b). Define load analysis of ship electrical system [2 marks]

c). List five electrical diagrams used in ship electrical system design. State their advantages. [7.5 marks]

# Paper 3 of 3 — MAR 314 MARINE ELECTRICAL TECHNOLOGY 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
COLLEGE OF ENGINEERING AND TECHNOLOGY  
DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: FIRST SEMESTER  
SESSION: 2024/2025  
DATE: 23/05/25



COURSE CODE: MAR 314  
COURSE TITLE: MARINE ELECTRICAL TECHNOLOGY  
TIME ALLOWED: 2HRS

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

1a). Discuss the notable four condition of a hand design that is influenced by Electrical Systems Onboard it. [4 marks]

b). With reference to ship electrical technology, explain ship primary essential services and ship secondary essential services. Outline the two examples of primary and secondary essential services. [4.5 marks]

2). In a container ship, two generators J and K are sharing load in parallel. The Real/True power, Reactive power and power factors of each generator is illustrated in the table below.

| Parameters      | Generator J | Generator K | Total bus bar Loading |
|-----------------|-------------|-------------|-----------------------|
| True/Real Power | 700 KW      | 650 KW      | ?                     |
| Apparent Power  | ?           | ?           | ?                     |
| Reactive Power  | ?           | 600 KVAR    | ?                     |
| Power Factor    | 0.7 lag     | ?           | ?                     |

Calculate the following: i). The reactive power of Generator J. [3.5 marks] ii). The apparent power of Generator J. [3.5 marks] iii). The apparent power of Generator K. [3.5 marks] iv). The power factor of Generator K. [3.5 marks] v). The total bus bar loading in KW, KVAR and power factor. [3.5 marks]

3). An oil tanker, has a 440V, 3-ph, induction motor. The induction motor has 6 poles and operates at 0.7 power factor lag and 6% slip. The motor drives a 15KW pump at 38.8 rev/s. The stator and mechanical losses are 750W and 480W respectively. Calculate:

i). The supply frequency. [4 marks] ii). The frequency of the rotor current. [4.5 marks] iii). The power input to the motor. [4.5 marks] iv). The current taken from the supply. [4.5 marks]

4a). In a war ship, a generator onboard has 60 conductors/ph and is needed to generate 500 volts/phase. Calculate the following: i). The needed flux per pole [2.5 marks] ii). The generator line voltage if the three stator winding are star connected. [2.5 marks]

b). A long shunt compound generator delivers a load current of 60A at 600V, has armature, series field and shunt field resistances of 0.07 ohms, 0.04 ohms and 350 ohms respectively. Calculate the generate voltage and the armature current. Allow 1V per brush for contact drop. [6 marks]

c). Describe in details, field, armature, prime mover, rotor, stator and slip rings as components of an AC generator, that is needed for onboard operations. [6.5 marks]

5a). Given that the generator E.M.F =  $2.22 \phi$  f z volts/ph. Use engineering judgement from first principle to prove it. [6.5 marks]

b). Explain theory of operations of an AC generator. Also explain Internal voltage drops due to the internal resistance and impedance of the generator, hysteresis and mechanical losses in AC generator. [5.5 marks]

c). Describe three conditions that must be met prior to synchronization of AC generators for offshore operations. [5.5 marks]

6a). Supposed a cargo ship needs 4 motors for various operations. Sketch/Draw a ship electrical system diagram that shows how 660V, 440V, 440V and 220V are supplied to various motors onboard the cargo ship. [8 marks]

b). Define load analysis of ship electrical system. [2 marks]

c). List five electrical diagrams used in ship electrical system design. State their advantages. [7.5 marks]